

List of Projects for Spring 2025

Course: Introduction To IOT

Project 1: Smart Irrigation System

Description:

A smart irrigation system is an automated solution designed to optimize plant watering based on real-time environmental data. It typically incorporates a network of sensors, actuators, and communication modules to achieve efficient water management. The objective is to develop a system that uses IoT technologies to monitor soil parameters and automate irrigation using a drip irrigation system.

Objectives

- Develop a system that can monitor soil moisture levels, soil temperature, humidity, and other parameters that affect the irrigation using sensors connected to an IoT network. Use the data collected by the sensors to determine the water requirements of the crops and optimize irrigation.
- Develop an automated drip irrigation system that can deliver water directly to the roots of the crops, reducing water usage and minimizing evaporation.
- Provide a live monitoring through an IOT platform
- At least 2 sensors should be used with at least one actuator.
- Failure analysis and calibration needs to be done.
- A prototype to demonstrate the proof of concept.

Data Collection Requirement:

Minimum Data collection requirement: 10 days

Project 2: Water Quality Monitoring System

A water quality monitoring system is a solution designed to continuously assess the condition of water by measuring various physical and chemical parameters. You are expected to develop a portable system that can be attached to a source of drinking water and monitor various parameters of the water quality. (Example: water coolers in hostels or mess kitchens)

Objectives:

- Utilize sensors to measure parameters such as pH level, turbidity, TDS and temperature of the water samples
- Implement a centralized dashboard for authorities to monitor drinking water quality data, receive alerts for abnormal conditions or contamination events, and manage water treatment processes.
- The system may be deployed in places where there are no Wi-Fi connections, so choose the communication module appropriately.
- Include features to view the history of data collected from the sensors and raise alerts when any current values are different from the trends.
- At least 2 sensors of two variants should be used with at least one actuator.
- Failure analysis and calibration needs to be done.

Data Collection Requirement:

Minimum Data collection requirement: 10 days

Project 3: Autonomous Wildlife Monitoring & Poaching Prevention

Description:

This project aims to create a simple and low-cost system to help protect wildlife from illegal poaching. The idea is to use sensors and wireless communication to detect suspicious activities in wildlife areas and alert forest officials in real-time.

Objectives

- Use sensors to detect any human movement in protected areas. By detecting movement, the system can immediately identify if someone is entering a restricted area where wildlife should be safe. This is the first step in spotting potential poachers.
- Integrate a sensor that listens for specific noises, such as the sound of gunshots or chainsaws. These specific sounds are strong indicators of illegal activities. The sensor acts as an extra check to confirm that the detected movement might be linked to poaching.
- When the sensors (movement or sound) detect something unusual, the system triggered to capture the image or record a short video clip as evidence of the activity helps law enforcement verify the event and take appropriate action.
- Use appropriate communication modules to send alerts and captured data to datacenter and forest officials so they can take immediate action.
- At least 2 sensors of two variants should be used with at least one actuator.
- Failure analysis and calibration needs to be done.

Project 4: Smart Railway Track Condition Monitoring with ML Integration

Railway track failures can lead to dangerous derailments, costly repairs, and service disruptions. Early detection of track faults is essential for ensuring safe and efficient railway operations. This project proposes an IoT-based system that employs multiple sensors and Machine Learning (ML) techniques to monitor railway track conditions in real time, predict potential failures, and alert authorities for prompt maintenance action.

2. Objectives

- The system should detect railway track faults such as cracks, gaps, and track damage using an array of sensors. The vibration, temperature, and ultrasonic sensors continuously collect real-time data on track conditions.
- Utilize ready-to-use Pre-trained ML models (trained on historical track failure data) to predict potential failures before they occur. The collected sensor data is fed into a Machine Learning model. This model, trained on historical track failure patterns, analyzes the sensor readings to identify anomalies and predict potential track failures.
- Provide real-time condition updates and trigger warning signals when immediate action is required. If the ML model or the sensor thresholds indicate a critical issue, the system activates a warning to alert nearby personnel immediately.
- The system transmits sensor data, alerts, and predictive insights to the OM2M cloud or any cloud platform. Railway authorities can then access this information remotely.

Data Collection Requirement:

Minimum Data collection requirement: 10 days

Project 5: Smart Patient Monitoring System:

This project aims to develop an IoT system that continuously monitors patient health and vital signs. The system is designed to provide early alerts and real-time control, ensuring that caregivers or healthcare professionals can respond quickly, even when patients are far from medical facilities.

Objective:

- Choose the appropriate sensors and IoT devices for monitoring patient health parameters. The sensors could include heart rate sensors, blood pressure monitors, temperature sensors, pulse oximeters, EEG etc.
- Process the patient data collected by the sensors and analyze it in real-time to identify any abnormal patterns or deviations from normal parameters.
- This can further be used to activate speakers/fans/dim lights etc.
- Develop an easy-to-use user interface for healthcare providers to access patient data and receive alerts and respond to emergency situations.
- At least 2 sensors of two variants should be used with at least one actuator.
- Failure analysis and calibration needs to be done.
- A prototype to demonstrate the proof of concept.

Data Collection Requirement:

Minimum Data collection requirement: 10 days

Project 6: Intelligent Parking System

Finding a parking spot in busy urban areas (such as malls) can be a frustrating experience for drivers, leading to increased traffic congestion, wasted time, and higher fuel consumption. Traditional parking systems are often outdated, and inefficient, and do not provide real-time information on available parking spaces. The problem statement for the IoT-based Intelligent Parking System project is to develop a system that uses IoT technologies to provide drivers with real-time information on available parking spots, optimize parking utilization and reduce traffic congestion.

Objectives:

- Develop a system that can monitor parking spots in real-time using sensors connected to an IoT network.
- Analyze the data collected by the sensors to identify available parking spots, parking patterns, and trends.
- Develop a user interface that can provide drivers with real-time information on available parking spots, including location, availability etc.
- Develop an intelligent parking allocation system that can optimize parking utilization, reduce traffic congestion, etc.
- Test and validate the system in a real-world environment to ensure its reliability and effectiveness in providing drivers with real-time parking information and reducing traffic congestion.
- Also, demonstrate the choice of parking slot with a suitable actuation.
- At least 2 sensors of two variants should be used with at least one actuator

Data Collection Requirement:

Minimum Data collection requirement: 10 days

Project 7: Analysis of Inflow and Outflow of People in a Closed Space Using Sensor Data

The objective of this experiment is to analyze the inflow and outflow of people inside a room or a confined space across different times of the day using sensor data. The experiment aims to transfer all collected data to cloud for further analysis and visualization. Key analyses include determining the most crowded and least crowded times of the day, identifying the time when the maximum number of people leave the room and detecting when the inflow is at its peak.

Objectives

- Plot a graph showing the inflow and outflow of people in the room over the entire duration of the experiment. This will provide an overview of the overall trends.
- Analyse the data to determine the most and least crowded time of the day by identifying the periods with the highest inflow/ outflow of people.
- Conduct additional relevant analyses based on the collected data to derive further insights into the movement patterns of individuals within the room.
- Display this data onto a web interface with your analysis
- At least 2 sensors of two variants should be used with at least one actuator.
- Failure analysis and calibration needs to be done.
- A prototype to demonstrate the proof of concept.

Data Collection Requirement:

Minimum Data collection requirement: 10 days

Project 8: Automated Home Security System

Enhancing home security through automation can provide peace of mind, deter break-ins, and offer immediate alerts to homeowners and authorities.

Objectives

- Use appropriate sensors for motion detection of door/window etc. monitor unauthorized entry.
- Integrate actuators to automatically lock/unlock doors or windows based on security protocols.
- Develop a web interface for real-time alerts and remote control of the security system.
- Simulate break-in scenarios to test the system's response and effectiveness.
- At least 2 sensors of two variants should be used with at least one actuator.
- Failure analysis and calibration needs to be done.
- A prototype to demonstrate the proof of concept.

Project 9: Personalized Climate Control System

Optimizing indoor climate can enhance comfort and energy efficiency in homes and offices.

Objectives

- Use temperature and humidity sensors to monitor indoor conditions.
- Implement actuators to mimic control of HVAC (Heating, Ventilation, and Air Conditioning) systems automatically.
- Develop a system to adjust indoor climate based on preferences and occupancy.
- Offer a web user interface for manual control and monitoring.
- Test the system for effectiveness in different room sizes and conditions.
- Display and analyze the data of the given room to configure the settings of your HVAC or similar system.
- At least 2 sensors of two variants should be used with at least two actuators.
- Failure analysis and calibration needs to be done.
- A prototype to demonstrate the proof of concept.

Data Collection Requirement:

Minimum Data collection requirement: 10 days

Project 10 Smart Home for the Aged

The aim of this project is to design and develop an IoT-based smart home system for the elderly to help them live independently and safely. The system should be able to monitor and control various devices and appliances in the home, such as lights, thermostats, doors, and security systems, using voice commands or a mobile application. The smart home system should be designed to address the needs of the elderly, who may have physical limitations or memory issues. The system should be easy to use and intuitive, with features that cater to their specific needs, such as automatic medication reminders and emergency alerts.

Objectives

- Automatic medication reminders and alerts
- Motion sensors to detect falls or other accidents. The sensors could include heart rate sensors, blood pressure monitors, temperature sensors, pulse oximeters, etc.
- Emergency alert system to notify caregivers or emergency services in case of an emergency with a suitable user interface.
- Identifying user patterns, and routines and identifying any anomalies.
- Remote monitoring and management of the home for caregivers or family members.
- Various scenarios need to be created, and data should be collected accordingly.
- At least 2 sensors of two variants should be used with at least one actuator.
- Failure analysis and calibration needs to be done.
- A prototype to demonstrate the proof of concept

Data Collection Requirement:

Minimum Data collection requirement: 10 days